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was conducted using TaqMan probes (Thermo Fisher Scientific, C_175677529_10) with the Bio-Rad CFX Opus 96 system. The genotyping process was carried out independently of clinical data and repeated in 10% of the samples as a quality control measure, showing a 100% agreement rate.

Statistical analyses were performed using R version 4.0.2, with a significance level of 0.05. Associations were assessed using Pearson's Chi-squared test or Fisher's exact test. Median age and days of hospitalization were compared

described in a previous systematic review (Araújo et al. 2022) using the PubMed database. A meta-analysis, combining odds ratios, was conducted using the metabin function of the meta package.

RESULTS

No significant differences or associations were found between outcomes and clinical variables, as presented in Table I. The Hardy-Weinberg equilibrium was observed in all groups, and no association between *IFITM3* rs12252 and severity or mortality was identified. The updated search for the systematic review did not yield any additional studies exploring the same outcome. Consequently, we combined our findings with three previous studies.(Alghamdi et al. 2021, Cuesta-Llavona et al. 2021). No association was observed for the allelic model (OR: 1.43, 95% CI: 0.99-2.07) (Figure 1a) and the C-recessive model

Table I. Evaluation of factors associated with severity and mortality outcomes. No significant effects were observed.

Phenotype	Mechanical ventilation		p-value	Death		p-value
	No, n = 83	Yes, n = 19		No, n = 64	Yes, n = 38	
Age (years)	60 (52,68)	65 (48,68)	0.997	60 (52, 67)	66 (51,70)	0.545
Hospitalization (days)	18 (9,35)	32 (10,60)	0.125	19 (9,41)	20 (10,35)	0.890
Sex female	40 (48%)	6 (32%)	0.189	33 (52%)	13 (34%)	0.089
Sex male	43 (52%)	13 (68%)		31 (48%)	25 (66%)	
Systemic arterial hypertension	46 (55%)	11 (58%)	0.845	37 (58%)	20 (53%)	0.610
Neurological diseases	17 (20%)	4 (21%)	>0.999	14 (22%)	7 (18%)	0.677
Hypothyroidism	9 (11%)	3 (16%)	0.692	7 (11%)	5 (13%)	0.758
Cancer	3 (3.6%)	0 (0%)	>0.999	1 (1.6%)	2 (5.3%)	0.554
Asthma	2 (2.4%)	2 (11%)	0.157	2 (3.1%)	2 (5.3%)	0.627
HIV infection	6 (7.2%)	1 (5.3%)	>0.999	5 (7.8%)	2 (5.3%)	>0.999
Alcoholism	4 (4.8%)	1 (5.3%)	>0.999	3 (4.7%)	2 (5.3%)	>0.999
Smoker	4 (4.8%)	1 (5.3%)	>0.999	4 (6.2%)	1 (2.6%)	0.648
rs12252 T/T	53 (64%)	11 (58%)	0.628	40 (62%)	24 (63%)	0.947
rs12252 T/C	30 (36%)	8 (42%)		24 (38%)	14 (37%)	

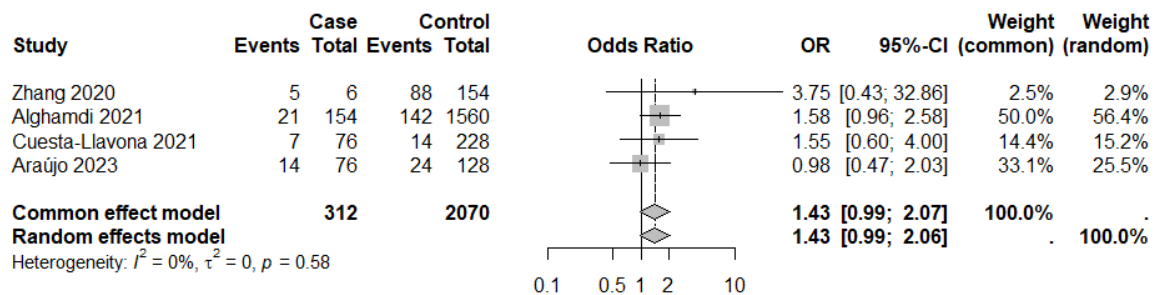
: 0.43-0.98), assuming non-ss the included studies ($p = 0.60$) (Figure 1c).

DISCUSSION

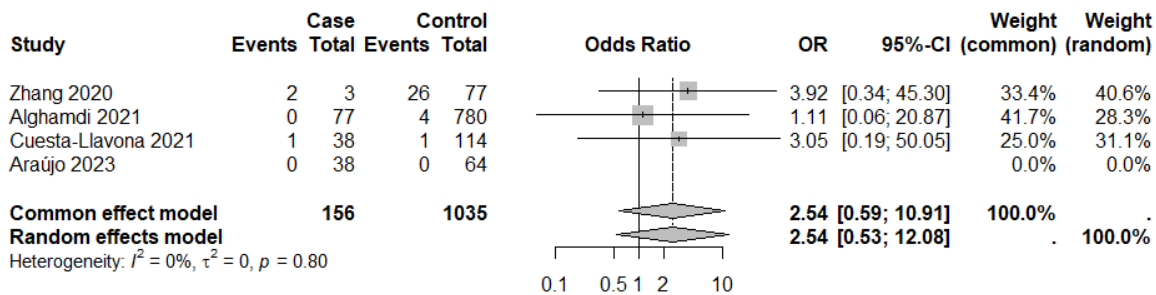
The association results of the rs12252 polymorphism with COVID-19 outcomes have

2020), Gomez and collaborators (Gómez et al. 2021), along with two others (Cuesta-Llavona et al. 2021, Schönfelder et al. 2021), did not identify increased odds of requiring intensive care unit treatment in CC carriers. However, the additive model revealed significance for the likelihood of hospitalization in Alghamdi and collaborators (Alghamdi et al. 2021) (OR: 1.65,

a



b



c

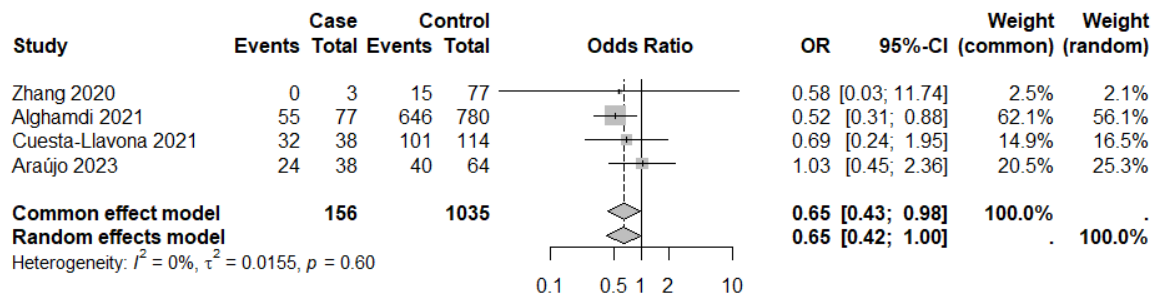


Figure 1. *IFITM3* rs12252 association with mortality under the (a) C-allele model. (b) T-recessive model. (c) C-recessive model. The T-recessive model was significant.

odds of death in TT carriers and CT subjects using meta-analysis. It is crucial to note that the T allele is rare, and its homozygosity was not found in our sample. Three meta-analyses explore the rs12252 polymorphism in susceptibility to SARS-CoV-2 infection, indicating the C allele as associated with an increased likelihood of infection (Gupta et al. 2022, Li et al. 2022, Pecoraro et al. 2023). Furthermore, in Pecoraro et al. a meta-analysis addressing death was conducted, observing no significance (Pecoraro et al. 2023). Consideration of the differences in the composition of compared groups in these studies is essential. The experimental design involves non-survivors versus survivors, and variations are observed within these groups. In our sample, all patients were hospitalized, whereas in Alghamdi et al, for instance, the survivor group included patients with mild to severe symptoms (Alghamdi et al. 2021). Two additional studies explored the likelihood of death in COVID-19 positive patients. Cuesta-Llavona and collaborators (Cuesta-Llavona et al. 2021) did not observe an association of the rs12252 polymorphism with the likelihood of death, while Alghamdi and collaborators (Alghamdi et al. 2021) reported significance in both the dominant model (OR: 1.76, 95% CI: 1.03-2.99, $p = 0.04$) and additive model (OR: 2.20, 95% CI: 1.16-4.20, $p = 0.01$). Replications in larger and ethnically diverse samples are warranted to further understand the role of *IFITM3* rs12252 in COVID-19 prognosis.

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Author contributions

All authors contributed to the study's conception. JLFA: conducted data curation, visualization, investigation, analysis, and original draft writing. VFB: conducted data curation, visualization, investigation, and analysis. LMB and REA: conducted data curation. RSA and LBR: Conceptualized the experimental design, provided supervision, obtained funding, and reviewed the manuscript. RPS: Conceptualized the experimental design, provided primary supervision, obtained funding, reviewed, and edited the manuscript. All authors read and approved the final manuscript.



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